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Estimating Regulation-Induced Substitution: The Effect of the Clean Air Act on Water and Ground Pollution

By MICHAEL GREENSTONE*

Recent research has demonstrated that the regulations imposed by the Clean Air Act Amendments' (CAAA's) are associated with reductions in manufacturing activity and improvements in ambient air quality (J. Vernon Henderson, 1996; Randy Becker and Henderson, 2000, 2001; Greenstone, 2002, 2003; Kenneth Chay and Greenstone, 2003a, b). Some portion of these reductions in emissions is due to lower output and changes in inputs. It is widely believed, however, that much of the reduction is due to air-pollution-abatement devices such as scrubbers and electrostatic precipitators. These abatement devices remove the pollutants from the air, but they are generally not eliminated, and the residuals are released into water bodies, landfills, or injected into the ground. The possibility that the CAAA's have the unintended consequence of increasing releases into other media has not been examined in the economics literature previously. If this cross-media substitution is important, then a complete accounting of the costs and benefits of this legislation requires measures of its extent and the damages associated with releases into each of these media.

This paper examines the effect of the CAAA's on the pollution emissions of the iron and steel industry (SIC codes 331 and 332) into all media during 1987–1997. The CAAA's divide U.S. counties into pollutant-specific nonattainment and attainment categories, based on ambient concentrations of the relevant pollutants. Emitters of the controlled pollutants in nonattainment counties are subject to substan-

tially greater regulatory oversight than are emitters in attainment counties. The iron and steel industry is chosen for this case study because it is an important source of industrial emissions of the pollutants regulated by the CAAA's.

I find that the lead, particulate matter (PM), and ground level ozone (O₃) nonattainment designations of the CAAA's are associated with reductions in total emissions of these pollutants in the iron and steel industry. This decline in total releases is accomplished through approximately equal reductions in releases into the air and into all other media. Thus, I find little evidence that the CAAA's increase releases into waterways and the ground. Although data on output are unavailable, there is suggestive evidence that emissions per unit of output declined.

I. Brief Background

A. Clean Air Regulations

The CAAA's direct the Environmental Protection Agency (EPA) to set separate national ambient air quality standards, a minimum level of air quality that all counties are required to meet, for six criteria pollutants, including lead, O₃, and PM. Based on the ambient concentrations, the EPA assigns separate nonattainment or attainment designations annually for each of the criteria pollutants in every U.S. county. The nonattainment designation is reserved for counties with ambient concentrations of the relevant pollutant that exceed the federal standard. Emitters of a pollutant in counties that are nonattainment for that pollutant are subject to stricter restrictions than emitters in attainment counties. Both state environmental agencies and the federal EPA are empowered to enforce the plant-specific regulations in these counties.¹ Previous

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¹ See Richard A. Liroff (1986) for a more detailed description of the CAAA's and the regulations they imposed on polluters.

research has demonstrated that the nonattainment designations are associated with substantial reductions in ambient air concentrations of O_3 and PM, as well as modest reductions in sulfur dioxide.

B. Emissions in Iron and Steel Production

The two predominant methods of steel and iron manufacturing in the United States are the Basic Oxygen Furnace (BOF) and the Electric Arc Furnace (EAF) processes. There are ample opportunities for the emissions of lead, PM, and volatile organic compounds (VOC's), a key ingredient in O_3 , into the air in both the BOF and EAF processes. In the BOF process, including coking but excluding desulfurization, exhaust gas generally contains between 16.5 and 37.4 kg of PM per ton of steel produced. In the cleaner EAF process, the level of PM in the exhaust gas averages approximately 11 kg per ton of steel. Depending on the lead content of the scrap metal used in production, lead accounts for 0.22–0.77 kg of the PM releases into the air from the EAF process. Approximately 3.3 kg of VOC's are created during the coking process for every ton of steel produced.²

Plants can reduce air releases by decreasing output or altering the production process so that unabated emissions per unit of output decline. The possible alterations to production include the burning of "cleaner" coal, use of higher-quality scrap metal as an input, and recycling and recovery of materials or waste for fuel use. These alternatives reduce emissions into all media.

An alternative approach is to use "end of pipe" abatement devices that reduce air emissions by separating the regulated pollutants from the escaping gas. Scrubbers, baghouses, and electrostatic precipitators are the most common air-pollution control devices in the iron and steel industries. Dry air or vapor scrubbers may be used to remove lead, PM, and VOC's from the exhaust gas stream. Baghouses and electrostatic precipitators are effective for the removal of PM and lead from the exhaust gas. The removal efficiency of these devices ranges from 90 to 99 percent, with scrubbers at the low end

of this range (J. Glynn Henry and Gary W. Heinke, 1996). However, the pollutants are not eliminated but instead converted into wastewater or solids, and these residuals are usually treated and then disposed of into landfills (either on- or off-site) or released into a water body. Consequently, these abatement devices reduce air emissions but increase releases into other media.

The subsequent analysis will test whether the nonattainment designation was associated with reductions in emissions into all media or only into the air. The results will lend insight into whether plants primarily relied on "end of pipe" devices to comply with the CAAA's.

II. Data Sources and Descriptive Statistics

A. Pollution Emissions Data

The *Toxics Release Inventory* (TRI) is one of the two primary data files used in this paper. The TRI is a plant-level data file that contains identifying information about the facility (e.g., name, county of location, industry), releases to air, water, land, and underground and transfers off-site of approximately 600 toxic chemicals. In the examined period (1987–1997), facilities were required to report to the EPA if they (i) had 10 or more full-time employees or the equivalent, (ii) were in a covered industry (the entire manufacturing sector is covered), and (iii) "manufactured" or "processed" more than 25,000 pounds or "otherwise used" more than 10,000 pounds of any listed chemical during a calendar year.³ Plants that meet (i) and (ii) must file separate reports for each toxic chemical that exceeds the threshold. The average iron and steel plant files reports for four TRI chemicals per year. Notably, the only penalties associated with the TRI are for false reports, not for high levels of emissions.⁴

³ A number of EPA documents provide more information on the precise definitions of "manufacture," "process," or "otherwise use" and on other important elements of the TRI. See, for example, the 1996 *Toxics Release Inventory: Public Data Release Report*, which is available online at (<http://www.epa.gov/tri/tridata/tri96/pdr/>).

⁴ The data on releases are generally estimated from engineering pollution production functions that take account of the inputs and the installed abatement devices. In a minority of the cases, the release data are due to monitoring.

² The estimates of emissions per ton of steel are from Richard Ackermann et al. (1999).

It is evident that the TRI is not a random sample of iron and steel plants. The number of iron and steel plants reporting per year ranges between 600 and 850. In a given year, the number of plants in the TRI is approximately one-third of the number of iron and steel plants reported to be operating in the *County Business Patterns* and roughly one-half of the plants with more than 10 employees in that data file. Beyond noting that it is likely that the TRI includes the plants that are the largest producers of pollution (and steel), there is little that can be done to account for the nonrandom sampling without information from another data set.⁵

The basis of the analysis is plant-level release data on the 270 toxic chemicals that are on the TRI list in each of the years from 1987 to 1997. Importantly, many of these toxic chemicals can be classified as either lead, PM, or VOC's, which are three of the six pollutants regulated by the CAAA's. This makes it possible to use the TRI for the first examination of the impacts of the CAAA's on plant emissions. Lead is one of the 270 listed chemicals. I contacted the EPA to determine which of the remaining chemicals are considered PM and/or VOC's, but these conversations did not lead to definitive classifications for many of the chemicals. Consequently, I also hired a Ph.D. chemist to assign the chemicals to these two categories. Ultimately, I assigned 75 chemicals to the PM category and 175 to the VOC category. Since 30 chemicals are included in both categories, I classify the remaining 50 chemicals as unregulated by the CAAA's.⁶ These 50 chemicals are henceforth referred to as the "unregulated" toxics. They may provide a valid counterfactual for the time path of media-specific releases of the regulated pollutants in the absence of the CAAA's.

The iron and steel industry was chosen for this case study, because it is one of the most important industrial sources of lead, PM, and VOC's. For instance, this industry accounts for 5.8 percent (18.9 percent) of emissions into the

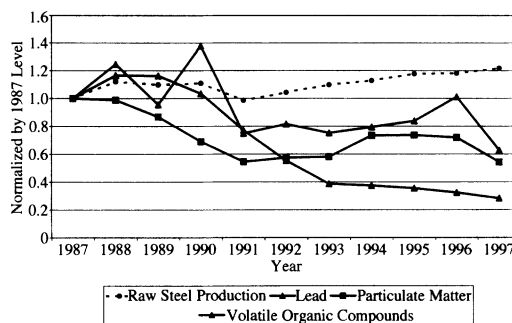


FIGURE 1. TRENDS IN RAW STEEL PRODUCTION AND AIR EMISSIONS OF LEAD, PARTICULATE MATTER, AND VOLATILE ORGANIC COMPOUNDS IN THE IRON AND STEEL INDUSTRY, 1987-1997

Notes: The data for raw steel production are taken from various issues of the American Iron and Steel Institute's *Annual Statistical Report*. The emission data are calculated from the *Toxics Releases Inventory (TRI)* data file. The steel-production data reflect the output of all steel plants, while the emissions data are based on the nonrandom sample of plants in the TRI. See the text for more details.

air (all media) of PM from the manufacturing sector according to calculations from the TRI data file. The comparable figures for VOC's and lead are 2.3 percent (2.3 percent) and 19.8 percent (13.5 percent), respectively.

Figure 1 reports trends in tons of raw steel produced by the entire iron and steel industry and emissions into the air of lead, PM, and VOC's from the plants in the TRI.⁷ Each series is normalized by its 1987 value. In 1987, total raw steel production was 89 million tons, and by 1997 it had increased more than 20 percent to almost 109 million tons. In this period of increased steel production, all three types of regulated air emissions declined dramatically. In particular, lead emissions were 37 percent lower in 1997 than in 1987, and PM and VOC's declined by 72 percent and 46 percent, respectively. These data suggest that air emissions of these pollutants per unit of output declined substantially during this period.

⁵ Although the TRI is a nonrandom sample, the selection rule that determined whether a plant was required to report was constant throughout the 1987-1997 period.

⁶ The categorization of the 270 toxics into the lead, PM, VOC, and "other" categories is available upon request from the author.

⁷ The raw steel production data are taken from various issues of the American Iron and Steel Institute's *Annual Statistical Report*. From this publication, it is not evident whether these production data cover all or part of SIC codes 331 and 332.

B. Nonattainment Data

The second primary data file contains the annual, attainment/nonattainment designations for each of the six regulated pollutants for the more than 3,000 U.S. counties. The *Code of Federal Regulations* (CFR) publishes the identity of these counties annually. These data were hand-entered from the CFR twice, and discordant entries were checked to ensure accuracy. Importantly, the TRI contains a county identifier, which allows the two files to be merged.

There are 490 counties with iron or steel plants in operation in the 1987–1997 period. All of these counties were attainment for lead from 1987 through 1991, but in the years from 1992 to 1996 between five and six counties were nonattainment. The number of O₃ nonattainment counties rises from 178 in 1987 to 197 in 1992–1993 and then declines to 146 in 1996.⁸ Finally, the number of PM nonattainment counties varied little in this period, ranging from 58 to 71.⁹

III. Econometric Framework and Results

In order to explore the effects of the nonattainment designations on the growth of emissions of iron and steel plants, the plant-level data are fit to the following equation:

(1) Percentage ΔE_{ipt}

$$\begin{aligned} &= (E_{ipt} - E_{ipt-1}) / [(E_{ipt} + E_{ipt-1}) / 2] \\ &= \sum_{t=88}^{97} \delta_t + \beta \text{Nonattain}_{pt-1} + \Delta \varepsilon_{ipt} \end{aligned}$$

⁸ There are separate nonattainment designations for O₃ and nitrogen dioxide, and in principle, a county could be nonattainment for one but not the other. Since O₃ is the result of a complicated chemical process that involves nitrogen dioxide and VOC's, I consider a county to be O₃ nonattainment if it is listed as nonattainment for either O₃ or nitrogen dioxide. All future references to O₃ nonattainment status refer to this combined measure.

⁹ During the 1987–1997 period, a county could be designated for either or both PM₁₀ (particles less than 10 μm in diameter) and total suspended particulates (particles less than 40 μm). In the analysis, a county is designated PM nonattainment if it is designated nonattainment for at least one of them.

where $\Delta \varepsilon_{ipt} = \alpha_p + \Delta u_{ipt}$. Here i indexes pollutant type (i.e., lead, PM, VOC's, or “unregulated”), p references a plant, t and $t - 1$ index the current and previous year, respectively. Percentage ΔE_{ipt} is the dependent variable and is measured as the percentage change between t and $t - 1$.¹⁰ The specification includes a full set of year indicators, δ_t . The sample size is 8,626 in all specifications.

The stochastic error term $\Delta \varepsilon_{ipt}$ includes a plant-specific component, α_p . The estimated variance-covariance matrix allows for dependence of observations within the same plant over time but assumes independence across plants.¹¹ Since there is little variation in nonattainment status over time, it is not feasible to estimate models with plant or county fixed effects without a dramatic loss in precision.

The parameter β captures the growth in emissions in nonattainment counties relative to attainment counties. Recall that nonattainment status is assigned at the county level. Henceforth, β is referred to as the “regulation effect.” Notably, the subsequent results are based on specifications that include the single “own-pollutant” nonattainment variable. For example, I make the restriction that lead nonattainment status has no effect on emissions of VOC's. These restrictions are discussed further below.

Table 1 reports the results from fitting equation (1) when total plant releases across all media is the dependent variables. The entries in the intersection of a column and panel report the parameter estimate, standard error, and R^2 value from a separate regression. Columns refer to a pollutant-specific nonattainment parameter, and panels are reserved for the pollutant-specific dependent variables. The low R^2 statistics reflect the difficulty in explaining the percentage change in plant-level emissions.

¹⁰ This measure of percentage change is an alternative to the difference of the natural logarithms of the levels in years t and $t - 1$. It is a second-order approximation to the log difference measure, ranges from -2.0 to $+2.0$, and portrays expansion and contraction symmetrically. Importantly, it allows the sample to contain observations on “entrants” and “exits,” that is, plants that operate in t but not $t - 1$ and those that operate in $t - 1$ but not t , respectively.

¹¹ An alternative is to allow for dependence of observations within a county by year. The standard errors on the regulation effects are essentially unchanged when this assumption is implemented.

TABLE 1—ESTIMATED REGRESSION MODELS FOR THE EFFECT OF NONATTAINMENT STATUS ON TOTAL RELEASE OF TOXICS, BY POLLUTANT TYPE

Percentage change in total releases of:	Nonattainment parameter		
	Lead	PM	O ₃
Lead	-0.071 (0.040)	—	—
R ² :	0.003	—	—
PM	—	-0.035 (0.018)	—
R ² :	—	0.006	—
VOC's	—	—	-0.056 (0.014)
R ² :	—	—	0.014
"Unregulated" toxics	-0.033 (0.032)	-0.022 (0.008)	-0.023 (0.006)
R ² :	0.003	0.004	0.004

Notes: Each set of entries is from a separate regression. The entries report the R^2 statistic and the parameter and standard error from the nonattainment variable indicated at the top of the columns. The dependent variables are the change in plant total (i.e., into all media) releases of lead (first panel), particulate matter (second panel), VOC's (third panel), and "unregulated" (fourth panel) toxics between a year and the previous year (i.e., between year t and $t - 1$), divided by the mean of the t and $t - 1$ levels. The data come from the 1987–1997 TRI sample of iron and steel firms, and the sample size is 8,626. All specifications include year fixed effects. The estimated variance–covariance matrix allows for dependence of observations within the same plant over time but assumes independence across plants. Standard errors are reported in parentheses. See the text for more details.

The first entry in the table suggests that, on average, total lead emissions by iron and steel plants declined by roughly 7.1 percent more in lead nonattainment counties than in attainment counties. The PM and O₃ parameters imply relative annual declines of 3.5 percent and 5.6 percent of PM and VOC releases in nonattainment counties, respectively, and both of these estimates would be judged different from zero by conventional criteria. Overall, these three parameters indicate that total releases of lead, PM, and VOC's declined by 21,000 pounds, 24.2 million pounds, and 12.2 million pounds, respectively, in the relevant nonattainment counties over the entire period.¹²

¹² The point estimates are smaller in magnitude but the qualitative story is similar when entrants or entrants and exiters are dropped from the sample.

The last panel presents the parameter estimates from separate regressions of total releases of the "unregulated" toxics on the lead, PM, and VOC nonattainment variables. A comparison of these entries with those in the first three panels reveals that the regulation effects on lead, PM, and VOC releases are of a larger magnitude. Under the assumption that the change in emissions of "unregulated" toxics is proportional to the change in output, these findings imply that nonattainment status is associated with reductions in emissions of the regulated pollutants per unit of output.¹³

Table 2 repeats this analysis for air releases (Table 2A) and releases into all other media (Table 2B; i.e., the sum of water releases, underground releases, and transfers to landfills). The air regulation effects reveal a negative association between each of the nonattainment designations and the relevant pollutant, although only the VOC regulation effect is statistically significant. If plants obtained these reductions in air pollution by installing "end of pipe" devices, then it is possible that releases to other media increased or at least decreased by less than air releases. However, the point estimates in Table 2B are at least as large in magnitude as those in Table 2A. This suggests that the regulations do not simply cause plants to substitute air releases for releases into other media; rather, emissions into all media decline. The regulation effects on the "unregulated" toxics are again smaller, in some cases substantially so, than the regulation effects on the pollutants regulated by the CAAA's.

Equation (1) is a parsimonious specification, which raises the possibility that the results in Tables 1 and 2 are not robust. I probed these results in a number of ways. For example, I experimented with specifications that separately add three-digit SIC by year indicators and state-by-year indicators. I also estimated equations that are weighted by the square root of the denominator of the dependent variable to account for differences in plant size. In all these cases, the qualitative results are unchanged.

¹³ These results can also be interpreted as providing a test of internal validity. From this perspective, it is reassuring that the regulation effects on the "unregulated" toxics are smaller than the regulation effects on the regulated pollutants.

TABLE 2—ESTIMATED REGRESSION MODELS FOR THE EFFECT OF NONATTAINMENT STATUS ON RELEASE OF TOXICS, BY POLLUTANT TYPE AND MEDIA

A. Air Releases			
Percentage change in air releases of:	Nonattainment parameter		
	Lead	PM	O ₃
Lead	-0.077 (0.048)	—	—
R ² :	0.002	—	—
PM	—	-0.024 (0.015)	—
R ² :	—	0.011	—
VOC's	—	—	-0.034 (0.014)
R ² :	—	—	0.015
"Unregulated" toxics	-0.038 (0.032)	-0.015 (0.007)	-0.017 (0.006)
R ² :	0.006	0.006	0.006
B. Releases into All Other Media			
Percentage change in other media releases of:	Nonattainment parameter		
	Lead	PM	O ₃
Lead	-0.083 (0.042)	—	—
R ² :	0.006	—	—
PM	—	-0.037 (0.016)	—
R ² :	—	0.004	—
VOC's	—	—	-0.040 (0.011)
R ² :	—	—	0.005
"Unregulated" toxics	0.013 (0.005)	-0.014 (0.006)	-0.017 (0.004)
R ² :	0.001	0.002	0.002

Notes: See the Notes to Table 1. In Part A, the dependent variables are the percentage changes in plant air releases of lead (first panel), particulate matter (second panel), VOC's (third panel), and "unregulated" toxics (fourth panel). In Part B, the dependent variables are the percentage changes in the pollutant-specific releases into all media besides air.

Another specification issue is whether the estimating equation should include the "cross-pollutant" regulation effects. The estimates of the "own-pollutant" regulation effects (e.g., the effect of PM nonattainment status on PM releases) are qualitatively similar to the ones described above when the "cross-pollutant" parameters are included. It is reasonable to suspect, however, that the regulation of one pollutant causes releases of another pollutant to

decline. In this case, it might be appropriate to treat the "cross-pollutant" parameters as genuine effects of regulation. Across the dependent variables, the "cross-pollutant" regulation effects are generally negative, small in magnitude, and statistically indistinguishable from zero. The notable exception is that VOC nonattainment status is associated with a statistically significant decline in lead and PM emissions into all media.

IV. Discussion and Future Directions

This case study of the iron and steel industries has demonstrated that the CAAA's lead, particulate matter, and ozone nonattainment designations are associated with reductions in total emissions of these pollutants in this industry. These declines in total releases are achieved through reductions in releases into the air and other media. Thus, I find little evidence that the CAAA's increase releases into waterways and the ground. Although data on output are unavailable in the TRI throughout the 1987–1997 period, there is suggestive evidence that emissions per unit of output declined.

There are at least two directions for future research. First, it is important to understand whether these results hold in other industries. The per-unit monetary value of emissions to regulated firms is a central, yet poorly understood policy parameter. This parameter is important, because it provides a direct measure of the costs to firms of mandated reductions in emissions and easily can be compared to the benefits of these reductions (Chay and Greenstone, 2003a, b, c). This paper's results suggest that nonattainment status may be a valid instrumental variable for total releases in plant-level production functions. Thus, the second direction for future research is to use the nonattainment variables as instruments to obtain estimates of the economic value of emissions to plants.

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